Lecture 14

The category of finite-dimensional representations

Objectives. *Lecture 13 classified the simple modules. We now study kernels, multiplicities, tensor products and duals of all finite-dimensional modules.*

By the end of this lecture you will be able to:

- Recover a module from its multiplicity spaces by evaluation.
- Construct tensor products and duals, with equivariant evaluation and coevaluation.
- Prove $L(\lambda)^* \cong L(-w_0\lambda)$ for $\lambda \in P_+$, and compute it for $\mathfrak{sl}_3$.
- Decompose $\mathbb{C}^3 \otimes (\mathbb{C}^3)^*$ under $\mathfrak{sl}_3$ and compute its endomorphism algebra.

14.1 Kernels, cokernels and Hom spaces

Throughout, $\mathfrak{g}$ is a complex semisimple Lie algebra. Fix a Cartan subalgebra $\mathfrak{h}$, positive roots R_+, simple roots $\alpha_1, \dots, \alpha_r$ and the Weyl group W. Write

$$Q_+ = \mathbb{Z}_{\geq 0}\alpha_1 + \cdots + \mathbb{Z}_{\geq 0}\alpha_r.$$

Let C be the category of finite-dimensional $\mathfrak{g}$-modules and intertwiners. Let Vect be the category of finite-dimensional complex vector spaces and linear maps. Let $F\colon C \to \mathrm{Vect}$ be the forgetful functor, which forgets the action.

Each $\mathrm{Hom}_{\mathfrak{g}}(V,W)$ is a subspace of $\mathrm{Hom}_{\mathbb{C}}(V,W)$, and composition is bilinear. The direct sum $V \oplus W$ has action $\rho_V(x) \oplus \rho_W(x)$. Its inclusions and projections are intertwiners, so it is a biproduct in C. The zero module is a zero object.

Proposition 14.1 (Abelian structure [P]). *Let $T\colon V \to W$ be an intertwiner.*

(i) The submodule $\ker T$, with inclusion ι, is a kernel of T in C.

(ii) The quotient module $W/\mathrm{im}\,T$, with quotient map q, is a cokernel of T in C.

(iii) The map $\bar{T}\colon V/\ker T \to \mathrm{im}\,T$, $v + \ker T \mapsto Tv$, is an isomorphism of modules.

Hence C is a $\mathbb{C}$-linear abelian category, and F is faithful and exact.

Proof. (i) Lecture 1 shows that $\ker T$ is a submodule. Let $S\colon U \to V$ be an intertwiner with $TS = 0$. Then $S(U) \subseteq \ker T$, so $S = \iota S'$ for a unique linear map S'. For $x \in \mathfrak{g}$ and $u \in U$,

$$\iota S'(xu) = S(xu) = xS(u) = \iota\big(xS'(u)\big).$$

Since ι is injective, S' is an intertwiner.

(ii) Let $S\colon W \to U$ be an intertwiner with $ST = 0$. Then S vanishes on $\operatorname{im} T$, so $S = \bar{S}q$ for a unique linear map $\bar{S}$. For $x \in \mathfrak{g}$ and $w \in W$, the map $\bar{S}$ intertwines:

$$\bar{S}\big(x(w + \operatorname{im} T)\big) = S(xw) = xS(w) = x\,\bar{S}(w + \operatorname{im} T).$$

(iii) The first isomorphism theorem makes $\bar{T}$ a linear bijection. It intertwines because $T(xv) = xTv$. Applying $\bar{T}^{-1}$ to $x\bar{T}(u) = \bar{T}(xu)$ shows that $\bar{T}^{-1}$ also intertwines.

Part (iii) identifies coimage with image, the remaining axiom of an abelian category. Kernels and cokernels are the vector-space ones, so F is exact. It is faithful because morphisms in $\mathcal{C}$ are linear maps. ∎

Semisimplicity of $\mathfrak{g}$ now enters. By Weyl's theorem (Lecture 5), every object of $\mathcal{C}$ is a finite direct sum of simple objects. By Lecture 13, the simple objects are, up to isomorphism, the $L(\lambda)$ with $\lambda \in P_+$, and $L(\lambda) \cong L(\mu)$ only if $\lambda = \mu$.

14.2 The isotypic decomposition

Definition 14.2 (Multiplicity space and evaluation)**.** For V in $\mathcal{C}$ and $\lambda \in P_+$, the *multiplicity space* is

$$M_\lambda(V) = \operatorname{Hom}_{\mathfrak{g}}\big(L(\lambda), V\big).$$

Let $\mathfrak{g}$ act on $M_\lambda(V) \otimes L(\lambda)$ through the second factor:

$$x(\varphi \otimes v) = \varphi \otimes xv.$$

The *evaluation map* is

$$\varepsilon_V\colon \bigoplus_{\lambda \in P_+} M_\lambda(V) \otimes L(\lambda) \longrightarrow V, \qquad \varphi \otimes v \longmapsto \varphi(v).$$

Theorem 14.3 (Isotypic decomposition [P])**.** *Let V be a finite-dimensional $\mathfrak{g}$-module.*

(a) Each $M_\lambda(V)$ is finite-dimensional, and $M_\lambda(V) \neq 0$ for at most $\dim V$ weights λ.

(b) The evaluation map ε_V is an isomorphism of $\mathfrak{g}$-modules.

(c) In every decomposition of V into simple modules, $L(\lambda)$ occurs exactly $\dim M_\lambda(V)$ times.

Proof. Step 1: finite support. Since $M_\lambda(V) \subseteq \operatorname{Hom}_{\mathbb{C}}(L(\lambda), V)$, it is finite-dimensional. Let $\varphi \in M_\lambda(V)$ be nonzero. The kernel of φ is a proper submodule of the simple module $L(\lambda)$, hence zero. For a highest vector v_λ of $L(\lambda)$, the vector $\varphi(v_\lambda)$ is nonzero and

$$H\varphi(v_\lambda) = \varphi(Hv_\lambda) = \lambda(H)\varphi(v_\lambda) \qquad (H \in \mathfrak{h}).$$

Simultaneous eigenvectors of $\mathfrak{h}$ with distinct eigenvalue functionals are linearly independent. Hence at most $\dim V$ weights λ have $M_\lambda(V) \neq 0$.

Step 2: equivariance and naturality. The rule $(\varphi, v) \mapsto \varphi(v)$ is bilinear, so ε_V is well defined. The map ε_V intertwines because $\varphi(xv) = x\varphi(v)$. For an intertwiner $T\colon V \to V'$, let $T_*\colon M_\lambda(V) \to M_\lambda(V')$ be $\varphi \mapsto T\varphi$. Then

$$T\varepsilon_V(\varphi \otimes v) = T\varphi(v) = \varepsilon_{V'}(T_*\varphi \otimes v).$$

If T is an isomorphism, so is T_*, with inverse $(T^{-1})_*$. Then ε_V is bijective exactly when $\varepsilon_{V'}$ is.

Step 3: simple modules. Let $V = L(\mu)$. Schur's lemma gives $M_\lambda(V) = 0$ for $\lambda \neq \mu$ and $M_\mu(V) = \mathbb{C}\,\mathrm{id}$. The source of ε_V is $\mathbb{C}\,\mathrm{id} \otimes L(\mu)$, and $\varepsilon_V(\mathrm{id} \otimes v) = v$. So ε_V is bijective.

Step 4: direct sums. Let $V = V_1 \oplus V_2$ with inclusions ι_k and projections p_k. Since $\iota_1 p_1 + \iota_2 p_2 = \mathrm{id}_V$, every $\varphi \in M_\lambda(V)$ satisfies

$$\varphi = \iota_1(p_1\varphi) + \iota_2(p_2\varphi), \qquad p_k\varphi \in M_\lambda(V_k).$$

The two terms are unique, since $p_k\iota_l = \delta_{kl}\,\mathrm{id}$. Thus

$$M_\lambda(V) = \iota_{1*}M_\lambda(V_1) \oplus \iota_{2*}M_\lambda(V_2).$$

Step 2 applied to ι_k gives

$$\varepsilon_V \circ (\iota_{k*} \otimes \mathrm{id}) = \iota_k \circ \varepsilon_{V_k} \qquad (k = 1, 2).$$

The maps ι_{k*} are injective. Hence the source of ε_V splits into the images of $\iota_{1*} \otimes \mathrm{id}$ and $\iota_{2*} \otimes \mathrm{id}$. On this splitting, ε_V becomes $\varepsilon_{V_1} \oplus \varepsilon_{V_2}$ followed by the isomorphism $\iota_1 + \iota_2 \colon V_1 \oplus V_2 \to V$. If ε_{V_1} and ε_{V_2} are bijective, so is ε_V.

Step 5: conclusion. The case $V = 0$ is immediate. By Weyl's theorem and Lecture 13, $V \cong L(\mu_1) \oplus \cdots \oplus L(\mu_k)$. Steps 3 and 4, with induction on k, make the evaluation map of the right side bijective. Step 2 transfers this to V, proving (b). Steps 2–4 also give

$$M_\lambda(V) \cong \bigoplus_{j=1}^{k} M_\lambda\big(L(\mu_j)\big), \qquad \dim M_\lambda(V) = \#\{j : \mu_j = \lambda\}.$$

This is (c). ∎

The image $V[\lambda]$ of $M_\lambda(V) \otimes L(\lambda)$ under ε_V is the canonical *λ-isotypic component*. Every simple submodule isomorphic to $L(\lambda)$ is the image of some $\varphi \in M_\lambda(V)$, so lies in $V[\lambda]$. Since ε_V is an isomorphism, this component is a direct sum of copies of $L(\lambda)$. Its splittings into simple submodules correspond to decompositions of $M_\lambda(V)$ into lines, by Schur's lemma.

14.3 Tensor products and duals

For modules (V, ρ_V) and (W, ρ_W) and for $x \in \mathfrak{g}$, define

$$\rho_{V\otimes W}(x) = \rho_V(x) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \rho_W(x).$$

Write $A = \rho_V(x)$, $A' = \rho_V(y)$, $B = \rho_W(x)$, $B' = \rho_W(y)$. The cross terms $A \otimes B'$ and $A' \otimes B$ occur in both products and cancel:

$$\begin{aligned}[\rho_{V\otimes W}(x), \rho_{V\otimes W}(y)] &= (AA' - A'A) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes (BB' - B'B) \\ &= \rho_{V\otimes W}([x, y]).\end{aligned}$$

The *tensor unit* $\mathbf{1}$ is the trivial module $\mathbb{C}$, on which every x acts by 0.

Proposition 14.4 (Tensor structure [P]). *If $S \colon V \to V'$ and $T \colon W \to W'$ are intertwiners, so is $S \otimes T$. The associativity map $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$, the unit maps $c \otimes v \mapsto cv$ and $v \otimes c \mapsto cv$, and the flip*

$$\sigma_{V,W} \colon V \otimes W \to W \otimes V, \qquad v \otimes w \mapsto w \otimes v,$$

are intertwiners. Moreover $\sigma_{W,V}\sigma_{V,W} = \mathrm{id}_{V\otimes W}$.

Proof. By linearity it suffices to check pure tensors. For $S \otimes T$,

$$(S \otimes T)(xv \otimes w + v \otimes xw) = xSv \otimes Tw + Sv \otimes xTw = x(Sv \otimes Tw).$$

On $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$, x acts as the sum of three terms, each applying x to one factor. The associativity map only moves brackets. The left unit map sends $x(c \otimes v) = c \otimes xv$ to $c\,xv = x(cv)$, and the right unit map is similar. For the flip,

$$\sigma(xv \otimes w + v \otimes xw) = w \otimes xv + xw \otimes v = x(w \otimes v).$$

The identity $\sigma_{W,V}\sigma_{V,W} = \mathrm{id}$ holds for vector spaces. ∎

The associativity, unit and flip maps are bijective, so their inverses also intertwine, by the argument used for $\bar{T}^{-1}$ in Proposition 14.1. Since F is faithful, a diagram of intertwiners commutes when its image under F does. Naturality and the pentagon, triangle and hexagon identities therefore hold because they hold in Vect. Hence C is symmetric monoidal, and F is a symmetric monoidal functor with $F(V \otimes W) = F(V) \otimes F(W)$.

For an operator A on V, the *dual map* is $A^{\mathsf{T}} : V^* \to V^*$, $\xi \mapsto \xi \circ A$. In a basis of V and its dual basis, the matrix of A^{T} is the transpose of the matrix of A. The *dual module* V^* has action

$$\rho_{V^*}(x)\xi = -\xi \circ \rho_V(x), \qquad D_{V^*}(x) = -D_V(x)^{\mathsf{T}}$$

in the dual basis. Transposition reverses products, so

$$\begin{aligned}[\rho_{V^*}(x), \rho_{V^*}(y)] &= \rho_V(x)^{\mathsf{T}}\rho_V(y)^{\mathsf{T}} - \rho_V(y)^{\mathsf{T}}\rho_V(x)^{\mathsf{T}} \\ &= -\big(\rho_V(x)\rho_V(y) - \rho_V(y)\rho_V(x)\big)^{\mathsf{T}} \\ &= \rho_{V^*}([x,y]).\end{aligned}$$

Lemma 14.5. *Let $\mathfrak{g}$ act on $\mathrm{End}_{\mathbb{C}}(V)$ by $x \cdot A = \rho_V(x)A - A\rho_V(x)$. The linear map*

$$\Psi : V \otimes V^* \to \mathrm{End}_{\mathbb{C}}(V), \qquad \Psi(v \otimes \xi)(u) = \xi(u)v,$$

is an isomorphism of modules.

Proof. For a basis (v_i) with dual basis (v^i), the operator $\Psi(v_i \otimes v^j)$ has the matrix unit E_{ij} as its matrix, so Ψ is bijective. For $x \in \mathfrak{g}$ and $u \in V$, the dual action gives $(x\xi)(u) = -\xi(xu)$. Hence

$$\begin{aligned}\Psi\big(x(v \otimes \xi)\big)(u) &= \xi(u)\,xv - \xi(xu)\,v \\ &= \big(\rho_V(x)\Psi(v \otimes \xi) - \Psi(v \otimes \xi)\rho_V(x)\big)(u).\end{aligned}$$

∎

Definition 14.6 (Evaluation and coevaluation)**.** With dual bases (v_i) and (v^i), define

$$\mathrm{ev}_V : V^* \otimes V \to \mathbf{1}, \;\; \xi \otimes v \mapsto \xi(v); \qquad \mathrm{coev}_V : \mathbf{1} \to V \otimes V^*, \;\; 1 \mapsto \sum_i v_i \otimes v^i.$$

Since $\mathrm{coev}_V(1) = \Psi^{-1}(\mathrm{id}_V)$, the coevaluation does not depend on the basis.

Proposition 14.7 (Rigidity [P])**.** *The maps ev_V and coev_V are intertwiners. With unit and associativity maps suppressed,*

$$(\mathrm{id}_V \otimes \mathrm{ev}_V)(\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V, \qquad (\mathrm{ev}_V \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^*}.$$

Proof. Since $\mathfrak{g}$ acts by 0 on the unit $\mathbf{1}$, we need $\mathrm{ev}_V \circ x = 0$ and $x\,\mathrm{coev}_V(1) = 0$. First,

$$\mathrm{ev}_V\big(x\xi \otimes v + \xi \otimes xv\big) = -\xi(xv) + \xi(xv) = 0.$$

Second, Ψ intertwines by Lemma 14.5, and $\Psi(\mathrm{coev}_V(1)) = \mathrm{id}_V$. Hence

$$\Psi\big(x\,\mathrm{coev}_V(1)\big) = \rho_V(x)\,\mathrm{id}_V - \mathrm{id}_V\,\rho_V(x) = 0,$$

and $x\,\mathrm{coev}_V(1) = 0$ because Ψ is injective.

For the zigzags, follow $v \in V$ and $\xi \in V^*$:

$$v \mapsto \textstyle\sum_i v_i \otimes v^i \otimes v \mapsto \sum_i v^i(v)\,v_i = v,$$
$$\xi \mapsto \textstyle\sum_i \xi \otimes v_i \otimes v^i \mapsto \sum_i \xi(v_i)\,v^i = \xi.$$

■

Proposition 14.7 makes V^* a left dual of V. In this convention, a left dual has its evaluation defined on $V^* \otimes V$. The module V^* is also a right dual, via the intertwiners $\mathrm{ev}_V\,\sigma_{V,V^*}$ and $\sigma_{V,V^*}\,\mathrm{coev}_V : 1 \mapsto \sum_i v^i \otimes v_i$. Their zigzags give $\sum_i v^i(v)\,v_i = v$ and $\sum_i \xi(v_i)\,v^i = \xi$.

The coproduct Δ and antipode S of Lecture 11 give

$$u(v \otimes w) = \Delta(u)(v \otimes w), \qquad (u\xi)(v) = \xi(S(u)v) \qquad (u \in U(\mathfrak{g})).$$

These formulas agree with the actions above on generators.

14.4 The dual of a simple module

Recall that $L(\lambda)$ is the direct sum of its weight spaces, with weights in $\lambda - Q_+$ (Lecture 12). Its set of weights $\mathrm{wt}\,L(\lambda)$ is W-invariant (Lecture 13).

Lemma 14.8 (Reversing the positive roots). *There is $w_0 \in W$ with $w_0(R_+) = -R_+$.*

Proof. The roots $-\alpha_1, \dots, -\alpha_r$ form a base of R, with positive roots $-R_+$. By transitivity of W on bases, cited in Lecture 9 [C: Humphreys, Section 10.3, Theorem, for a root system], some w_0 maps the first base onto the second. Since w_0 is linear and permutes R, it maps the roots that are nonnegative combinations of the first base onto those of the second. That is, $w_0(R_+) = -R_+$. ■

Theorem 14.9 (Dual highest weight [P]). *For $\lambda \in P_+$,*

$$L(\lambda)^* \cong L(-w_0\lambda).$$

Proof. Put $V = L(\lambda)$.

Step 1: weights of V^*. Choose a basis (v_k) of weight vectors, $v_k \in V_{\mu_k}$, with dual basis (ξ^k). For $H \in \mathfrak{h}$,

$$(H\xi^k)(v_l) = -\xi^k(Hv_l) = -\mu_l(H)\delta_{kl} = -\mu_k(H)\,\xi^k(v_l).$$

So $H\xi^k = -\mu_k(H)\xi^k$, and $\mathrm{wt}\,V^* = -\mathrm{wt}\,V$.

Step 2: V^* is simple. Let $U \subseteq V^*$ be a submodule. The annihilator $U^\perp \subseteq V$ of U is a submodule. Indeed, if $v \in U^\perp$ and $\xi \in U$, then $x\xi \in U$ and

$$\xi(xv) = -(x\xi)(v) = 0.$$

Since $\dim U^\perp = \dim V - \dim U$ and V is simple, $U = 0$ or $U = V^*$. By Lecture 13, $V^* \cong L(\nu)$ for some $\nu \in P_+$.

Step 3: the lowest weight of V. Let $\mu \in \operatorname{wt} V$. Then $w_0^{-1}\mu \in \operatorname{wt} V$, so $w_0^{-1}\mu = \lambda - \beta$ with $\beta \in Q_+$. Hence

$$\mu = w_0\lambda - w_0\beta \in w_0\lambda + Q_+,$$

because $-w_0$ maps R_+, and hence Q_+, into Q_+. Also $w_0\lambda \in \operatorname{wt} V$.

Step 4: comparison. By Steps 1 and 3, $\operatorname{wt} V^* \subseteq -w_0\lambda - Q_+$ and contains $-w_0\lambda$. Also $\operatorname{wt} L(\nu) \subseteq \nu - Q_+$ and contains ν. Thus $\nu + w_0\lambda$ and $-(\nu + w_0\lambda)$ both lie in Q_+. Linear independence of the simple roots gives $Q_+ \cap (-Q_+) = 0$, so $\nu = -w_0\lambda$. ∎

The proof uses only $w_0 \in W$ and $w_0(R_+) = -R_+$. Since $L(\nu)$ determines ν, the weight $-w_0\lambda$ does not depend on the choice of w_0.

For $\mathfrak{sl}_3$ with the diagonal Cartan subalgebra $\mathfrak{h}$, let $\varepsilon_i \in \mathfrak{h}^*$ be given by $\varepsilon_i(H) = a_i$ for $H = \operatorname{diag}(a_1, a_2, a_3)$, so $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$ on $\mathfrak{h}$. Then $R_+ = \{\varepsilon_i - \varepsilon_j : i < j\}$, with simple roots $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$. The reflection in the root $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$, with coroot $H_{13} = E_{11} - E_{33}$, is

$$s_{\alpha_1+\alpha_2}(\mu) = \mu - \mu(H_{13})\,(\varepsilon_1 - \varepsilon_3).$$

It exchanges ε_1 and ε_3 and fixes ε_2. Hence it sends $\varepsilon_i - \varepsilon_j$ to $\varepsilon_{4-i} - \varepsilon_{4-j}$, a negative root when $i < j$. So $w_0 = s_{\alpha_1+\alpha_2}$ satisfies $w_0(R_+) = -R_+$.

Lecture 13 identified the defining module $\mathbb{C}^3$ with $L(\omega_1)$, whose highest vector is the first standard basis vector e_1. Here $\omega_1 = \varepsilon_1$. For the other fundamental weight $\omega_2 = \varepsilon_1 + \varepsilon_2 = -\varepsilon_3$,

$$-w_0\omega_1 = -\varepsilon_3 = \omega_2, \qquad (\mathbb{C}^3)^* \cong L(\omega_2).$$

14.5 The module $\mathbb{C}^3 \otimes (\mathbb{C}^3)^*$

Let $V = \mathbb{C}^3$ be the defining $\mathfrak{sl}_3$-module, with basis (e_i) and dual basis (e^i). The map Ψ of Lemma 14.5 identifies $V \otimes V^*$ with $\mathfrak{gl}_3(\mathbb{C})$, where $e_i \otimes e^j \mapsto E_{ij}$ and

$$x \cdot A = [x, A].$$

Under Ψ, the composite $\operatorname{ev}_V \sigma_{V,V^*}$ sends $v \otimes \xi$ to $\xi(v) = \operatorname{tr} \Psi(v \otimes \xi)$. So this composite is the trace. The coevaluation sends 1 to $\sum_i e_i \otimes e^i$, which Ψ sends to I_3. Hence

$$\operatorname{ev}_V \sigma_{V,V^*} \operatorname{coev}_V(1) = \sum_{i=1}^{3} e^i(e_i) = 3 = \dim V.$$

Define the *trace projector* and its complement, transported to $\mathfrak{gl}_3(\mathbb{C})$ by Ψ:

$$P = \frac{1}{3} \operatorname{coev}_V \operatorname{ev}_V \sigma_{V,V^*}, \qquad P(A) = \frac{\operatorname{tr} A}{3} I_3, \qquad Q(A) = A - \frac{\operatorname{tr} A}{3} I_3.$$

Both maps are intertwiners. Indeed, P is a composite of intertwiners, and $Q = \mathrm{id} - P$. The value 3 computed above gives

$$P^2 = \frac{1}{9} \operatorname{coev}_V \big(\operatorname{ev}_V \sigma \operatorname{coev}_V\big) \operatorname{ev}_V \sigma = \frac{3}{9} \operatorname{coev}_V \operatorname{ev}_V \sigma = P.$$

Since $Q = \mathrm{id} - P$, it follows that

$$Q^2 = \mathrm{id} - 2P + P^2 = Q, \qquad PQ = QP = P - P^2 = 0.$$

So $V \otimes V^* = \operatorname{im} P \oplus \operatorname{im} Q$ as modules, with

$$\operatorname{im} P = \mathbb{C} I_3 \cong L(0), \qquad \operatorname{im} Q = \ker \operatorname{tr} = \mathfrak{sl}_3.$$

On $\mathfrak{sl}_3$ the action is the adjoint action. Submodules of the adjoint module are ideals (Lecture 1), and $\mathfrak{sl}_3$ is simple (Lecture 10). So $\operatorname{im} Q$ is simple. For $H = \operatorname{diag}(a_1, a_2, a_3)$ in $\mathfrak{h}$, the vector $e_1 \otimes e^3$, which corresponds to E_{13}, satisfies

$$\begin{aligned}[E_{12}, E_{13}] &= E_{12}E_{13} - E_{13}E_{12} = 0 - 0 = 0,\\ [E_{23}, E_{13}] &= E_{23}E_{13} - E_{13}E_{23} = 0 - 0 = 0,\\ [H, E_{13}] &= (a_1 - a_3)E_{13}.\end{aligned}$$

The root vectors E_{12} and E_{23} generate $\mathfrak{n}_+$, since $[E_{12}, E_{23}] = E_{13}$. So E_{13} is a highest vector of weight $\varepsilon_1 - \varepsilon_3 = \omega_1 + \omega_2 = \alpha_1 + \alpha_2$. Therefore

$$\mathbb{C}^3 \otimes (\mathbb{C}^3)^* \cong L(0) \oplus L(\omega_1 + \omega_2), \qquad 9 = 1 + 8.$$

Endomorphism algebra. Let $T \in \operatorname{End}_{\mathfrak{g}}(\mathfrak{gl}_3)$. With respect to $\mathfrak{gl}_3 = \mathbb{C}I_3 \oplus \mathfrak{sl}_3$, write

$$T = \begin{pmatrix} T_{00} & T_{01} \\ T_{10} & T_{11} \end{pmatrix},$$

where $T_{10} \colon \mathbb{C}I_3 \to \mathfrak{sl}_3$ and $T_{01} \colon \mathfrak{sl}_3 \to \mathbb{C}I_3$. Each block is T composed with an inclusion and with P or Q, so it is an intertwiner.

The simple modules $\mathbb{C}I_3$ and $\mathfrak{sl}_3$ have dimensions $1 \neq 8$, so Schur's lemma gives $T_{10} = T_{01} = 0$, $T_{00} = a$ and $T_{11} = b$ id for some $a, b \in \mathbb{C}$. Thus $T = aP + bQ$, and

$$\operatorname{End}_{\mathfrak{g}}\bigl(\mathbb{C}^3 \otimes (\mathbb{C}^3)^*\bigr) = \mathbb{C}P \oplus \mathbb{C}Q \cong \mathbb{C} \times \mathbb{C}.$$

Both isotypic components are simple. By the description of isotypic components, $\mathbb{C}I_3 \oplus \mathfrak{sl}_3$ is the unique decomposition into simple submodules.

14.6 The structure of the category

A *rigid symmetric tensor category* is a $\mathbb{C}$-linear abelian symmetric monoidal category with finite-dimensional Hom spaces and objects of finite length, in which $(S, T) \mapsto S \otimes T$ is bilinear, $\operatorname{End}(\mathbf{1}) = \mathbb{C}$, and every object has a left and a right dual.

Theorem 14.10 (Structure of C). *Let $\mathfrak{g}$ be complex semisimple. Then C is a $\mathbb{C}$-linear semisimple abelian rigid symmetric tensor category with $\operatorname{End}_{\mathfrak{g}}(\mathbf{1}) = \mathbb{C}$. The functor F is faithful, exact and tensor. If $\mathfrak{g} \neq 0$, then C has infinitely many isomorphism classes of simple objects.*

Proof. Propositions 14.1, 14.4 and 14.7 and the right duals give the stated structures. Weyl's theorem gives semisimplicity. Hom spaces are finite-dimensional. Dimensions bound the length of strict chains of submodules. Bilinearity of $(S, T) \mapsto S \otimes T$ holds for linear maps, hence for intertwiners. Every linear map $\mathbb{C} \to \mathbb{C}$ intertwines, so $\operatorname{End}_{\mathfrak{g}}(\mathbf{1}) = \mathbb{C}$. If $\mathfrak{g} \neq 0$, then $R \neq \emptyset$ and $r \geq 1$. The weights $m\omega_1$, $m \geq 0$, are distinct and dominant, so the modules $L(m\omega_1)$ are pairwise nonisomorphic. ∎

If the zero algebra is admitted as semisimple, its category is Vect, with one simple object $\mathbb{C}$.

Part II studies the multiplicities $\dim M_\nu(L(\lambda) \otimes L(\mu))$ of simple modules in tensor products.

Self-checks

(1) Identify $L(2\omega_1 + \omega_2)^*$ for $\mathfrak{sl}_3$.

(2) Let $V = L(\lambda) \oplus L(\lambda) \oplus L(\mu)$ with $\lambda \neq \mu$. Find $\dim M_\lambda(V)$. Is the splitting of $V[\lambda]$ canonical?

(3) Why does the dual action carry a minus sign?

Answers. (1) $-w_0\omega_1 = \omega_2$ and $-w_0\omega_2 = w_0\varepsilon_3 = \varepsilon_1 = \omega_1$, so $L(2\omega_1+\omega_2)^* \cong L(\omega_1+2\omega_2)$. (2) Two. No: the diagonal $\{(v, v, 0)\}$ is another simple submodule of $V[\lambda]$. (3) Transposition reverses brackets. The sign restores the bracket rule and makes ev_V equivariant.

Summary

Let $\mathfrak{g}$ be complex semisimple. Evaluation identifies each finite-dimensional module V with $\bigoplus_\lambda M_\lambda(V) \otimes L(\lambda)$, with finitely many nonzero terms. Evaluation, coevaluation and the flip are intertwiners. Hence C is a semisimple rigid symmetric tensor category. For $\lambda \in P_+$, the dual of $L(\lambda)$ is $L(-w_0\lambda)$.

Reading

A. Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Sections 4.2 and 8.1. P. Etingof, *Lie Groups and Lie Algebras* (arXiv:2201.09397), the sections on tensor products and dual representations. J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 10.3, for the imported transitivity of W on bases.